DO NOW — RECALL

# Processors, scheduling and LMC practice

Name: _____ Class: _____ Date: _____

**1. What does LMC STA do?**

**2. What is the role of MAR?**

**3. What is a time quantum?**

**4. Full adder 1+1+0: sum and carry bits?**

DO NOW — CHECK / OBJECTIVES

# Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

### 1. Model answer

Copies ACC contents to the specified mailbox.

### 2. Model answer

Address of the memory location being accessed.

### 3. Model answer

Maximum CPU slice before RR rescheduling.

### 4. Model answer

$10_2$: sum0, carry1.

### By the end, you can…

• Trace the LMC accumulator and mailbox separately: INP overwrites ACC, while STA copies to storage.

• Schedule independent CPU jobs with an explicit arrival/time rule and label every Gantt boundary.

• Distinguish processor speed, memory cache and interrupt/OS services when explaining performance.

### Vocabulary — know these terms

ACC — Accumulator register.

FCFS — First Come First Served.

SJF — Shortest Job First.

pipeline — Overlap instruction stages.

NEW LEARNING

# Read, annotate and explain

### Understand

Trace the LMC accumulator and mailbox separately: INP overwrites ACC, while STA copies to storage.

### Apply

Schedule independent CPU jobs with an explicit arrival/time rule and label every Gantt boundary.

### Evaluate

Distinguish processor speed, memory cache and interrupt/OS services when explaining performance.

**Explain one step in your own words, using the correct technical terms.**

I DO — WORKED EXAMPLE

# Follow the worked method

### Trace sequential instruction fetch when PC=90.

### Predict first

PC holds address of next instruction, not the instruction contents.

### Explain and check

Explain the reason for the result.

WE DO — GUIDED PRACTICE

# Try → Discuss → Improve

YOU DO — INDEPENDENT PRACTICE

# Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

### Question 1

Using LMC code above, trace ACC after each input, LDA, SUB and OUT. [6]

### Question 2

Processes A (burst 4), B (burst 2) and C (burst 1) all arrive at t=0. For FCFS, use the stated arrival-list order A, B, C to break ties; for non-preemptive SJF choose the shortest available job. Compare execution orders and mean waiting times. [6]

YOU DO — INDEPENDENT PRACTICE

# Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

### Question 3

Evaluate clock speed, cache and pipelining for a task with heavy memory latency and branches. [6]

EXIT TICKET / REFLECTION

# Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

**LMC STA X loads ACC from mailbox X.**

**How confident are you? Circle one: Confident / Getting there / Need help**

Explain your choice, identifying one concept you can now use independently.

## GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

Next step: revisit the model or a relevant revision-guide section if needed.